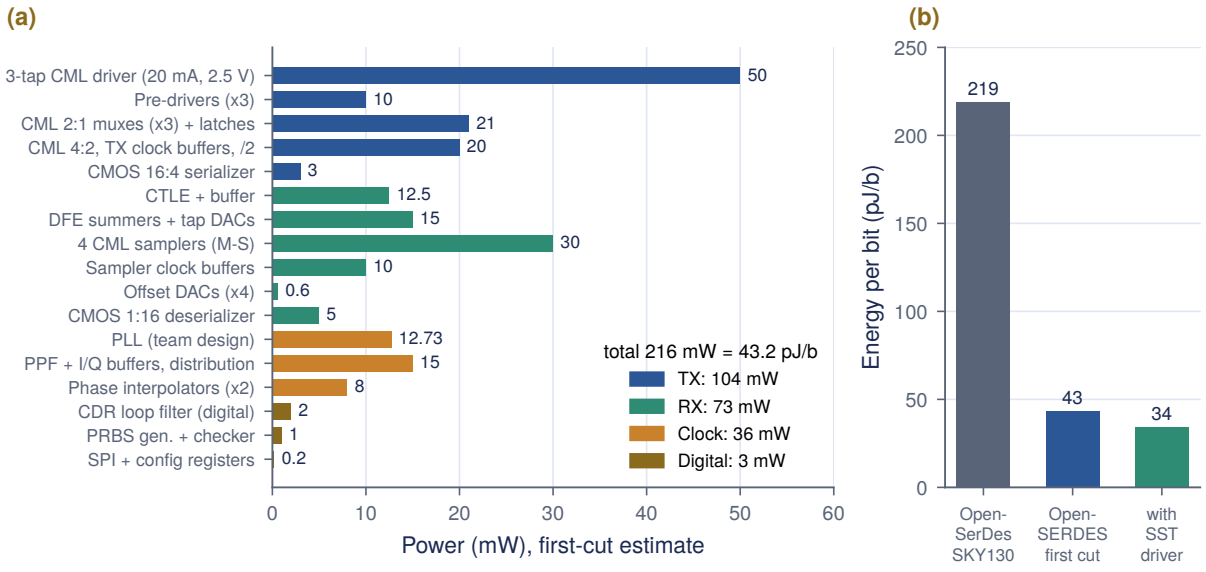
(a)
3-tap CML driver (20 mA, 2.5 V)
Pre-drivers (x3)
CML 2:1 muxes (x3) + latches
CML 4:2, TX clock buffers, /2
CMOS 16:4 serializer
CTLE + buffer
DFE summers + tap DACs
4 CML samplers (M-S)
Sampler clock buffers
Offset DACs (x4)
CMOS 1:16 deserializer
PLL (team design)
PPF + I/Q buffers, distribution
Phase interpolators (x2)
CDR loop filter (digital)
PRBS gen. + checker
SPI + config registers
50
10
21
20
3
12.5
15
30
10
0.6
5
12.73
15
8
2
1
0.2
0
10
20
30
40
50
60
Power (mW), first-cut estimate
total 216 mW = 43.2 pJ/b
TX: 104 mW
RX: 73 mW
Clock: 36 mW
Digital: 3 mW
(b)
Energy per bit (pJ/b)
250
200
150
100
50
0
219
43
34
Open-SerDes SKY130
Open-SERDES first cut
with SST driver